Communication Challenges in the Everyday Lives of Deaf People in India

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One-sixth of the global Deaf population resides in India, constituting over 63 million people. Despite the large number of Deaf in India, research on the Deaf population is mostly concentrated in the Global North. In this qualitative and quantitative study, we interview 15 Deaf participants and survey 131 members of the Indian Deaf community in order to define the challenges faced by them on a day-to-day basis. We particularly focus on challenges faced at the workplace, due to the large influence of employment on quality of life. Our work reveals key challenges faced by the Deaf community due to the inaccessible nature of Indian workspaces, and contrasts these experiences with that of the Global North. We also cover common challenges faced in other everyday scenarios and evaluate the various workarounds used by the Deaf community today. Finally, we propose inclusive design recommendations for various applications commonly used by the community.

CCS Concepts: • **Human-centered computing → Empirical studies in accessibility**; *Accessibility design and evaluation methods*; **User studies**.

Additional Key Words and Phrases: Accessibility, Individuals with Disabilities & Assistive Technologies, Workplaces, Empirical study that tells us about people

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1 INTRODUCTION

The Deaf and Hard of Hearing (DHH) community constitutes over 5% of the global population [4]. For the members of this community, hearing loss limits communication ability and poses distinct challenges in their everyday lives. Most of the accessibility research that studies these challenges are based in the Global North[1]. These studies explore the socioeconomic context of the lives of the Deaf, such as the role of family [34], communication challenges in educational institutions [51], and access to health care [44]. Prior work in resource-rich countries has also explored challenges faced by Deaf employees in a workplace setting—examining communication patterns, career barriers and difficulties faced by the DHH community in the Australian workplace [70], communication preferences of Deaf professionals in supervisory roles in the USA [36], and experiences of stress and fatigue among the Deaf at work [68]. Other studies assume well-resourced settings and explore technological interventions like custom camera setups to recognize sign language [3, 16], use of video/telecommunications relay services to make calls [8, 79], and teletypewriter technology on smartphones for emergency calls [82].

Many of these solutions and insights do not generalize to a country like India due to the large differences in cultural context, access to technology and employment opportunities. One example of this cultural difference is the nature of

[1]Global North refers to the countries located primarily in the northern hemisphere that have historically been identified as "the West" or "first world" due to perceptions of their relative wealth, technology, and global dominance.

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Indian Sign Language (ISL). Unlike American or British Sign Language, ISL is not a *single* language. It has a variety of dialects, such as Delhi Sign Language, Kolkata Sign Language, Bangalore-Chennai Sign Language, and Bombay Sign Language [1]. This diversity has impeded standardization and makes communication amongst the Deaf community hard. The structure of ISL is also exclusive—the ISL lexicon has a high level of iconicity and compounding[2] not seen in other sign languages [33].

Other factors specific to India include accessibility-related cultural taboos (reported in work on the Indian blind population [45, 47]), limited opportunities for education in mainstream schools [45], and the lack of trained interpreters [48]. The Indian Deaf population also warrants its own study due to its large size—India is home to 63 million Deaf people or one-sixth of the global Deaf population [28, 37]. Additionally, hearing loss is the second most common disability in India, representing 18.9% of the disabled population [62]. Despite the above, a formal study on the challenges faced and potential support systems for the Deaf community in India has not received much research attention.

In this work, we seek to address this gap by asking the research question: *"What are the key communication challenges faced by the working Indian Deaf community today?"* We focus on communication as it is a fundamental challenge faced by the Deaf community. Additionally, we focus on the workplace due to the importance of employment for earning a livelihood, developing skills, and collaborating with others [52]. The DHH community has historically experienced higher rates of unemployment and lower levels of education compared to people without hearing loss [69]. The unemployment rate of the community is a staggering 73.9%[3], underlining the importance of such a study.

In order to achieve our goal, we conduct both qualitative and quantitative studies of the Indian Deaf population. Our qualitative study consists of 15 virtual interviews with employed members of the Deaf population, facilitated by a certified sign language interpreter. Our survey consists of 131 responses to an online form filled by employed members of the Deaf community. Through these methods, we study the challenges faced by the Indian Deaf population at work and in their everyday lives. We find that the Deaf population relies on simplistic workarounds, such as writing or using basic ISL, to navigate through a variety of critical conversations. These workarounds are often ineffective, forcing the community to rely on hearing people to help them carry out everyday tasks and thereby reducing their agency. Assistive technological solutions have either not been built or have not yet been adopted at scale by the community.

This work makes several contributions. First, we discuss the problems arising due to the stigma associated with sign language education, the grammatical inconsistencies in the English used by the Deaf population, and the diversity of ISL. Next, we examine in detail the challenges faced by the Deaf community at the workplace and other everyday situations. Lastly, we contrast these challenges with those experienced by the employed American Deaf population, and propose several inclusive design recommendations for applications commonly used by the Indian Deaf community.

2 RELATED WORK

In this section, we examine existing work in two related areas—global work on DHH-related accessibility and accessibility work in the Indian context. The latter has primarily focused on the visually impaired community, along with a few initial studies on ISL.

2.1 Research on the DHH Community across the Globe

Hearing loss severely impairs an individual's ability to communicate with other people. Several technological solutions exist for the DHH community to communicate with hearing people, such as video/telephone relay services [8, 9],

[2] An example of compounding in ISL is the word 'brother', which is signed by compounding the signs for 'male' and 'sibling'.
[3] https://yourstory.com/2016/12/hearing-disabilities-india/amp

video chat [6], Internet Captioned Telephone Services [7], Assistive Listening Devices [2], and Instant Messaging. The methods used by the DHH community to communicate with hearing people have been well studied [36, 39, 69] and uncover several communication-related workarounds including avoidance, lip reading, optimizing volume of speech, confirming the message, attracting the attention of the hearing person, and using simpler words and signs.

Communication challenges have also been studied in specific contexts. Significant work has been done in the context of education given its importance in shaping the lives of the Deaf, such as the experiences of students with hearing impairment in educational institutions [51, 63], hearing students' perspectives on the inclusion of DHH students [25, 42], and methods to teach and assess DHH students [23, 24, 57, 60]. These studies have found that DHH students may not always disclose their needs and often settle for sub-par accommodations [22, 51]. They also dismantle misconceptions that technology can remove all access barriers or that technology built for one individual can generalize to others [51].

Another context that has been studied extensively is the home. The home plays a sensitive role in shaping the experience of Deaf children, since more than 90% of Deaf children are born to hearing parents [5]. Hearing parents of Deaf children have to make several decisions regarding communication and language choices on behalf of their children. Research on the factors that influence this decision-making [31, 43] reveal that parents learn to view hearing loss in one of two ways: the *sociocultural* view of deafness aligning with the use of sign language, or the *audiological* view aligning with the use of the family's spoken language. The efficacy of these methods depends on the characteristics of each child. However, it was observed that parents often chose a method based on inadequate information obtained from professionals and the priorities they had for their children [81].

Lastly, an analysis of the workplace context is crucial due to its potential to support accessibility interventions. Several studies chronicle the current experiences of the DHH community in the workplace, and propose and evaluate approaches to reduce the challenges faced by the community [36, 58, 61, 67, 68, 70]. Quantitative findings from a survey-based study indicate that the most difficult workplace situations were meetings, training activities, and work-related social functions [70]. There exist multiple accommodations to address these challenges, such as better lighting and furniture rearrangement to facilitate seeing other people better, and access to sign language interpreters [58]. Apart from workplace provided accommodations, it is apparent that certain personal attributes like persistence and self-advocacy skills, as well as the presence of supportive and helpful co-workers are important to facilitate job satisfaction and success among DHH people [32]. There has also been research on the awareness of the Americans with Disabilities Act (ADA)[4] [78] among employers, and the perspective of employers regarding the employment of DHH people [40]. However, none of these studies focus on India, while our work focuses on the working Indian DHH community.

2.2 Accessibility Research in India

Most of the above cited work on the DHH community has been localized to the Global North, where both the resources and awareness to address these challenges are relatively higher. In the wider space of accessibility research, significant work has been done over the past decade on the visually impaired community in India. This body of research has studied various aspects of visually impaired people's lives such as navigation [49, 50], education [46, 47], technology adoption and has proposed solutions [53, 71, 80]. Beyond the visually impaired community, there has also been research on developmental disabilities such as autism [13, 17, 73], cerebral palsy [75], speech and motor impairment [30]. There is limited work on the DHH community in India. Existing work has primarily focused on educating Deaf children [62, 83], methodologies to teach English to DHH students [35, 66], and teachers' perspectives on the education of Deaf

[4]ADA is a comprehensive civil rights law in USA prohibiting discrimination based on disability in employment, state and local government programs, public accommodations, commercial facilities, transportation, and telecommunications.

children [65]. We did not find any existing research that studies the broader set of challenges faced by the Deaf in India, or that specifically focuses on the workplace context.

With respect to the Indian DHH community, the main focus has been on studying Indian Sign Language (ISL) [76, 77]. These studies focus on the diversity of ISL and its unique characteristics that differ from other sign languages and the many spoken languages of India. They examine the several regional dialects of ISL used across India [1], and its high iconicity and use of compound signs [33]. Another crucial feature of ISL that has been examined is its variation with class differences. Members of the educated middle-class Deaf community use a unified and relatively standardized sign language, while the rural Deaf have no exposure to this urban form. Instead, the rural Deaf use organically evolved *home-sign systems* [72], specialized to the socio-linguistic context of their communities. The Indian Sign Language Research and Training Centre[5] (ISLRTC) was established in 2015 to further research on ISL and promote its standardisation and usage. There have also been recent efforts by the government of India to standardise ISL, starting with the introduction of an ISL dictionary in 2018 [11], and the National Education Policy 2020 which states that Indian Sign Language will be standardised and taught in schools [10].

In summary, the challenges of the DHH community have mainly been studied in the Global North. However, many of these insights and technological solutions do not directly generalize to a country like India due to several reasons: (a) lower access to resources and education, (b) lack of accommodations for DHH employees and (c) distinctive features of ISL.

3 METHOD

We conducted an online survey of the challenges faced by the DHH community in India, followed by semi-structured video interviews. Both the survey and the interviews had a focus on the workplace context. The study was approved by the Institutional Review Board at <Anonymous Institute>. In this section, we present our survey and interview methodologies, along with our data collection and analyses techniques.

3.1 Survey

Our survey consisted of 18 questions—12 multiple-choice questions and 6 open-ended questions—and was administered via Google Forms. The survey was organised in three sections with questions on (a) communication challenges and workarounds (such as "*Where do you find it difficult to communicate with hearing people?*" and "*Which apps do you use to talk to your family and friends?*" (b) workplace context (such as "*Does your workplace have a sign language interpreter?*" and "*How do you communicate with your coworkers?*") , and (c) demographic information. The survey was administered in English. All the survey questions were short, simple, and unambiguous to make them accessible to participants with low English proficiency. The surveys were piloted with two researchers and a certified ISL interpreter who provided feedback on question framing. The survey was accompanied with a video of a certified ISL interpreter explaining the purpose of the survey in ISL.

The survey form was distributed by a certified ISL interpreter, an NGO for Deaf, and a philanthropic organization supporting work on the Deaf within their networks using WhatsApp and email. Furthermore, we asked the survey participants to share it within their WhatsApp groups. In order to be included in the survey, participants needed to be deaf or hard of hearing, currently living in India, and be employed currently or in the past. The final optional question of the survey allowed participants to volunteer for the interviews, in which case they provided their email ID and phone

[5]http://www.islrtc.nic.in/

number. No Personally Identifiable Information (PII) was collected, unless the participants volunteered to participate in the interviews.

In total, we received 131 valid responses (107 male, 24 female, age=27.9±5.9 years) over 30 days spanning Jan–Feb 2021. All the survey respondents self-attested as Deaf or hard of hearing, out of which 47.3% of respondents have profound hearing loss (*i.e.*, complete deafness). Thirty-three participants have graduate degrees, 65 have undergraduate degrees, and the rest have a high school diploma.

We open-coded the responses to the open-ended questions and analysed them using inductive interpretive coding [64]. We identified themes such as 'access to interpreters at workplace', 'usage of WhatsApp', and 'challenges in everyday scenarios' (like banking, hospital visits). These emerging themes informed areas of focus in our semi-structured interviews in the next phase of the study. As a result, the Findings section of the paper predominantly draws on the interviews for quotes. We will specifically indicate that we are reporting survey data where relevant.

3.2 Interview

We conducted semi-structured interviews after the survey during Feb–Aug 2021. Participants were recruited either from survey respondents who volunteered to participate in the interview, or with the help of a sign language interpreter who shared the interview request with her friends and colleagues. In total, we interviewed 15 participants (8 male, 7 female, age=21–34 years). All the participants were either currently or previously employed and lived in urban areas. Summarized information about the participants is available in Table 1.

In the interviews, we asked participants about their sign language education and proficiency, preferred mode of communication with hearing people in their family or with friends, coworkers and strangers, and workplace accessibility challenges and workarounds. Next, we asked them to recall recent difficulties faced by them in a public place or workplace and the approach they used to handle the situation. At the end of the interview, participants were invited to share open comments and express any generic concerns.

All interviews were conducted remotely by the first two authors using the Google Meet video conferencing tool, in the presence of a certified ISL interpreter (female, 27 years old). All the calls were video-recorded with the consent of the participants. Participants were informed that the data would only be used for research purposes, and that no PII would be shared. The interviews lasted 30–75 minutes. The authors and the ISL interpreter interacted in English, while the ISL interpreter and the participant communicated in Indian Sign Language. When quoting participants, we use the exact translation provided by the interpreter. The interviews were transcribed soon after they were conducted. All participants' names have been anonymized in this paper. Participants were paid 500 INR for their participation.

We analyzed the interview transcripts using an inductive and iterative approach. All authors periodically discussed the interview data to identify emerging themes. We used an inductive, interpretive coding approach on this data [64]. Initial themes included 'communication with family', 'challenges concerning phone calls', 'lip-reading', and 'challenges in hospital'. We iterated over the data to produce higher-level themes including 'ISL education', 'work related challenges', and 'communication workarounds', and abstracted them out to the five main themes presented in this paper.

4 FINDINGS

In this section, we will discuss findings from our survey and interviews. First, we set the context of ISL—explaining the state of sign language education in India, the role ISL plays in the way the Indian deaf population perceives English and its various forms. Next, we analyse our participants' occupations—discussing sectors that employ them, details of their

Table 1. Demographic details of the interview participants

ID	Age	Sex	Hearing Loss Level	Location	Highest Education Degree	Sector	Occupation
P1	26	M	Mild	Hyderabad	Under-graduate	IT/BPO	Movie editor
P2	26	F	Moderate	Coimbatore	Under-graduate	IT/BPO	Security Guard
P3	37	M	Moderate	Bombay	High School	Education	Teacher
P4	34	M	Moderate	Hyderabad	Under-graduate	IT/BPO	Expenditure Auditor
P5	25	M	Profound	Theni	Post-graduate	Retail	Data Entry Operator
P6	25	M	Profound	Coimbatore	Under-graduate	E-commerce	Warehouse Assistant
P7	29	M	Profound	Coimbatore	Post-graduate	IT/BPO	Process Executive
P8	21	F	Profound	Hyderabad	Under-graduate	NGO	Video Creator
P9	25	F	Profound	Hyderabad	Post-graduate	IT/BPO	Customer Support
P10	24	F	Profound	Hyderabad	Under-graduate	IT/BPO	Expenditure Auditor
P11	32	F	Profound	Delhi	Under-graduate	Education	Teacher
P12	33	F	Profound	Hyderabad	Under-graduate	IT/BPO	Expenditure Auditor
P13	26	M	Moderate	Trivandrum	Under-graduate	Education	Teacher
P14	25	M	Moderate	Trivandrum	Under-graduate	IT/BPO	Software Developer
P15	21	F	Profound	Hyderabad	Under-graduate	NA	Currently Unemployed

jobs, and accessibility-related challenges in their workplaces. Lastly, we examine other everyday situations (such as shopping or using public transportation) and detail communication issues faced in each of these places.

4.1 Sign Language Education

In developed countries, children with deafness attend mainstream schools (also known as integrated/inclusive schools) with trained educators [14]. In India, sign language education has been formally recognised very recently (in July 2021 [12]). Hence, a majority of our participants (11) attended schools for Deaf. The other 4 participants attended mainstream schools and struggled with social stigma throughout.

The participants who attended mainstream schools said that they did so as their parents did not want them to be seen as "different". They recalled that they were given a hearing aid at school and asked to sit in the front, in an attempt to ensure that they could hear and lipread the teachers.

> "I studied with other hearing students near my house. I went to NISH for college where I learnt sign language. It was difficult for me while studying in school. I sat in the first row in class. I used hearing aids and could understand. Sometimes I could not understand some words and kept asking the teacher. If it gets cold like during the rainy season, then my hearing reduces and I could not hear anything at all." – *P14*

This approach resulted in negative experiences—P2 and P12 did not understand their teachers at all and relied solely on notes written by their friends, while P14 stated that he faced social isolation in school as a result of his disability.

Among the participants who attended schools for the Deaf, P1 mentioned that he was explicitly forbidden from using sign language at school. He recalled that he was not allowed to converse with teachers in sign language and was encouraged to lipread and speak instead. Similarly, P10 mentioned:

> "I can read lips a little, but I mostly use ISL. I don't speak... When I was a child, the teacher used to teach me and force me to speak. I went to a deaf school where there was a speech therapist, but I could not understand him." – *P10*

This encouragement to speak is widely prevalent in India, as we found that 49.6% of our survey respondents have undergone speech therapy. P4 shared that he was encouraged to "*practice speaking slowly and clearly*" by his parents, as they feared that he would be socially isolated if he relied only on sign language. Despite this, there was an overwhelming consensus amongst our participants that they preferred sign language to speech and that they were more comfortable with using ISL.

Five of our participants did not learn ISL at home or school but picked it up informally from peers, by watching videos on YouTube, or through apps like the DEF-ISL app[6]. Between the 3 methods, our participants found peer learning to be the the most effective. Three participants mentioned clearing doubts by asking their friends and said that practising with friends gave them the confidence needed to use sign language in public. These participants later learnt ISL formally in college as a part of their undergraduate studies or by completing a sign language diploma course.

4.2 English literacy

When communicating with the hearing community, the use of ISL is often not feasible due to the lack of interpreters. India has only 301 certified sign language interpreters [48] for a Deaf population of 63 million [37]. Moreover, these interpreters are very expensive (costing 13.7–20.5 USD/hour). Nine of our participants reported having access to interpreters only in their offices, while the others did not have access to them anywhere. Thus, written English became the dominant method used while communicating with the hearing community. However, our participants face various communication challenges while writing and reading English due to linguistic differences between ISL and English.

The first and perhaps most apparent difference between ISL and English is the underlying sentence structure. While English follows the SVO (subject - verb - object) order for arranging words in a sentence, ISL follows the SOV (subject - object - verb) order [76]. Due to this, the Deaf community tends to follow the SOV order even while communicating in English, resulting in sentences that are confusing to read. A similar phenomenon has been observed in other countries as well. For instance, Italian Sign Language has an underlying SOV structure whereas the Italian language follows the SVO order, which results in non-native signers of Italian Sign Language following the SVO order while signing [26]. The other major difference between ISL and English is the absence of articles (like 'a' or 'the') and connectives (like 'and' or 'or') in ISL [18]. This results in missing articles/connectives when the Deaf write in English too. Below are a few examples of the unique sentence structure used by the Indian Deaf community, taken from WhatsApp conversations with our participants:

> "I go start my exercise walking now it" instead of "I shall now start my walking exercise".
> "He is Venkatesan exercise time short so lazy" instead of "He is Venkatesan, who's exercise time is short and hence he is so lazy."
> "Before I wash shirt and jeans" instead of "I washed my shirt and jeans before this".

Finally, P13 (a teacher) noted that Deaf students often had difficulty in understanding idioms such as "*time flies*".

4.3 Diversity of ISL

As covered previously in Section 2.2, ISL is not a single, standardised sign language. This variation in ISL became a frequent topic of conversation during our interviews as our interpreter (who is from Hyderabad) would sometimes fail to understand signs used by our participants who were from different parts of India. An example of this variation is the follows:

[6]https://play.google.com/store/apps/details?id=in.eightfolds.deafenabled

> "The sign for 'marriage' is shown by a mangalsutra (an auspicious necklace used in Hindu weddings) *[performs a sign that draws out a necklace]* in Chennai, while in Hyderabad it is shown by the holding of hands *[touches right hand fingers with left]* .. just like when the bride and groom hold hands during Hindu weddings. Similarly in North [India] the sign for mother is *[touches her nose]* which indicates the nose ring while in Hyderabad you need to do two signs for mother - the North Indian sign followed by this *[holds fist to cheek]*." – *Interpreter*

We observed that these variations in sign language are reflective of India's cultural and linguistic diversity. India is comprised of 29 culturally-diverse states and is home to 184 languages (each of which is spoken by more than 10,000 speakers) [38]. ISL, just like spoken languages, is strongly influenced by the culture of its signers. Three of our participants mentioned knowing multiple signs for a given word, as they had grown up in multiple cities. P9 stated that even signs for basic words like '*mother*' and the days of the week varied across the different regions she grew up in.

Apart from the inconvenience, participants reported instances where these variations in ISL caused problems during lectures, training workshops, and workplace meetings.

> "Sometimes she [my manager at work who knows sign language] will stop and say 'I can't understand what you are signing.' She'll ask me twice or thrice again. She won't understand as she is from North [India]. Sometimes I get fed up and just text her to explain." – *P1*.

Apart from the diversity of ISL itself, 4 participants also reported the mixing of American Sign Language (ASL) with ISL. This influence of ASL on ISL was observed in participants from tier-1 cities like Delhi and Bombay. This phenomenon is similar to code-switching, wherein a speaker alternates between 2 or more languages in the same conversation or utterance [15]. In particular, we found the ASL-ISL switch similar to the Hindi-English switch (called *Hinglish*), which is commonly observed among Hindi-speaking Indians [55]. Our participants specified that one key reason for code-switching is that ASL has a unique advantage. Unlike the two-handed ISL, all ASL alphabets can be signed using one hand [27], thus enabling them to sign while holding objects with the other hand. All 4 participants mentioned using ASL mainly for WhatsApp video calls as they could hold their phone with one hand and use the other hand for signing. While these participants used ASL for such scenarios, all of them reported that they were not fluent in ASL and could only fingerspell the alphabets and a few other basic signs. Other participants who did not use ASL would sign by using their body or face as a substitute for one hand while holding objects.

4.4 Work

In our survey, 53.4% (70 out of 131) of respondents stated that they faced workplace related communication challenges. Thus, it emerged as the key challenge faced by the Deaf community (Figure 1). Below, we discuss industries that employ the Deaf community, what the job entails, and the day-to-day challenges faced by our participants.

4.4.1 Job profiles and work settings. Our survey was filled by participants working at 53 different companies. To gain a deeper understanding of sectors that employ the Deaf community, we mapped these 53 companies to 14 sectors (Figure 2). We found that the IT/BPO and Finance sectors are the most common sectors that employ people with deafness. We note that this data may not be an accurate representation across India and across all sectors as the survey was distributed using snowball sampling with the initial participants from the IT/BPO sector. Our interview participants worked in 6 of these sectors, with a majority of them (8) employed by the IT/BPO sector.

Our survey participants in the IT/BPO sector mainly worked as software engineers (13.7%), expense auditors (13.0%), and data entry operators (6.1%). P4 described his job as an expense auditor:

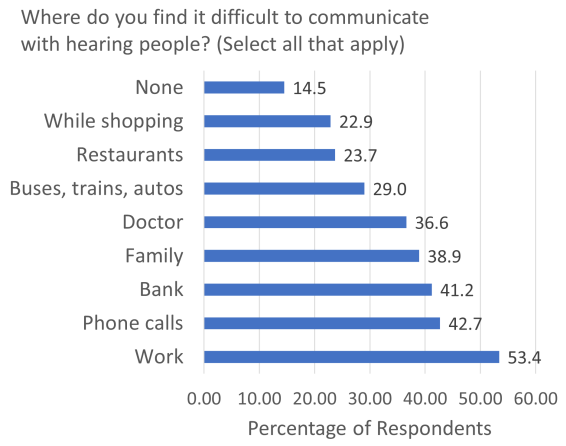

Fig. 1. Survey results for scenarios where Deaf find it difficult to communicate

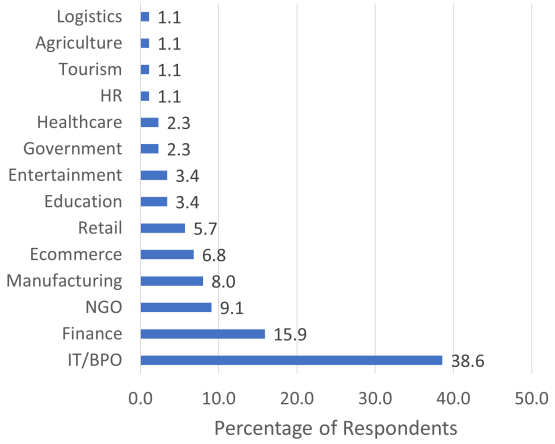

Fig. 2. Sectors that survey respondents worked in

> "There is an app called Concur. We approve bills there. Bills come from third party, and we see if the amount being claimed matches the amount on the bill. If the amount is not matching, we have to manually delete it and enter the amount printed on the bill. Then we get to the next bill. We have to check each bill within 10 seconds, otherwise our reports per hour goes down. We have to reach [at least] 97% of the weekly target... They measure the bottom performer of each week. We have to login for 8 hours per day, with 1 hour of break." – *P4*

We found that such job descriptions were typical as they were designed to minimise communication with other colleagues or customers, and worked around the English literacy issues that the Deaf community has.

To further understand the type of work done by our survey respondents, we categorised each job into 2 classes: white collar and blue collar jobs. White collar jobs focus on knowledge work with opportunities for hierarchical advancement [59]. E.g., software engineers, teachers, and accountants. Blue collar jobs are physically labor intensive with stationary hierarchical level (typically in low-ranked positions) [59]. E.g., data entry operators, warehouse workers at e-commerce companies, and restaurant staff. We found that 61.4% of our survey respondents work in blue collar jobs and 38.6% in white collar jobs. The IT/BPO and finance sectors are the major employers of white collar jobs, while manufacturing and e-commerce sectors mainly employed blue collar workers. Due to COVID-19, a majority of our survey participants (59.4%) were working from home, 21.0% were going to their workplace daily, and the remaining 19.6% followed a hybrid working model (going office 2-3 times every week).

Finally, we noted that despite already having a college degree, 10 of our interview participants completed additional training courses to get employed. These courses are conducted by NGOs, who teach skills like writing in English, typing on keyboard, using Microsoft Excel, and video editing. Our participants found these courses to be valuable—it taught them the in-demand skills and connected them with prospective employers. Thus, these additional courses significantly increase their likelihood of getting employed.

4.4.2 Access to Interpreters at Workplace. Only 52.2% of our survey respondents had a full-time sign language interpreter in their workplace, 37.7% didn't have access to any interpreter, and the rest had access to interpreter only during crucial

meetings and/or large events. Even among our interview participants, 6 participants stated that their office lacks an interpreter. Despite the dearth of interpreters, 77.9% of our participants prominently used ISL at their workplace. This was made possible using interesting workarounds. E.g.,

> "We don't have any interpreter. Initially we faced difficulty in talking to the supervisor but later we started teaching them sign language. We started with alphabets and a few words. Now they can sign! There is also another deaf girl who can talk, which makes things easier" – *P2*

All our interview participants, and 99.2% of our survey respondents work as Individual Contributors—roles wherein an individual can contribute to the growth of the organisation with minimal coordination with other colleagues. The lack of Deaf people in management roles is reflective of the communication challenges they face in workplace settings. P6 confirmed this by stating he was unable to grow in leadership roles in his organization due to communication challenges. However, a few of our survey respondents (6.1%) worked in organizations run by the Deaf community where they did not face such challenges. Among our interview participants, P8 worked in an NGO consisting mainly of Deaf employees.

4.4.3 Meetings. Meetings are a crucial component of office life, with employees across organisations attending meetings for an average of 6 hours/week [56]. Effective communication in meetings is key for a successful corporate career. Although 99.2% of our survey respondents worked in Individual Contributor roles, 61.8% of them stated that they attend one or more meetings every week, thus underlining the importance of meetings. These meetings comprise of weekly performance discussions for each employee, rewards for good performance, or training for future assignments. Despite the large number of meetings, only 52.6% of our survey respondents agreed that they are able to talk to others during workplace meetings. To understand this better, we asked our interview participants about their experience in such meetings.

We found Deaf teams with an interpreter are more successful at conducting meetings. In the organisations of P4, P10 and 12, the interpreter jointly leads the team with another manager. Thus, the interpreter has both work context and is comfortable with sign language. In case of P9's organization, the interpreter does not have a management role. She conducts a separate follow-up meeting with the Deaf employees after a meeting concludes, to reiterate everything that was discussed.

Other participants (like P5) who did not have access to interpreters reported that they were unable to participate in meeting discussions. P5 would mostly skip meetings and rely on meeting minutes prepared by the team leader. While this workaround allowed Deaf participants to be updated about the discussion, it precluded them from meaningfully contributing.

A few participants use lip reading to follow meeting discussions. For instance, P7 uses hearing aids to attend meetings. Having learnt lip reading and received speech therapy during his childhood, P7 can listen and respond to others as conversations happen during a meeting. However, this approach could get frustrating, as P7 described:

> "I feel bad when others are speaking fast and I am unable to tell them to slow down. If someone is talking to me fast and I am unable to understand then I ask someone else to explain what that person is saying. That person usually summarizes it instead of telling me the whole story and treats me like a baby... I don't like that. I sometimes turn off my video and sit because I don't understand what people are saying." – *P7*

Five participants mentioned trying the auto-generated captions feature offered by video calling applications (like Google Meet). While they found the feature useful, they stated 2 critical drawbacks that prevented them from using

live captions regularly. First, participants struggled to read and understand captions when someone spoke fast; *"If a person is speaking too fast... only 50% of the time I can understand, rest of it I miss." – P1.* The other drawback was that the caption generator often make errors in understanding Indian accents and pronunciations, especially for Indian names.

4.4.4 Conversations with managers. Communication with one's manager is vital to success at work. In some offices with interpreters, the interpreter acted as the team lead for Deaf employees. Although the employees had another reporting manager, their single point of contact was the interpreter, who resolved the Deaf employees' doubts and raised concerns with the reporting manager as required. Communication between Deaf employees and hearing managers is challenging in offices without interpreters. The default mode of communication is exchanging written messages (on a piece of paper, or over email/WhatsApp). While writing enables limited communication, it has two drawbacks. First, it is slow and becomes impractical for long, two-way conversations. Second, participants were concerned that writing about an issue "*appears excessively formal*" and may be taken out of context as a complaint against the manager. E.g., P6 described his hesitation with writing an email requesting promotion:

> "Initially, my [Deaf] seniors helped me understand and do my role. Now I want to go to the next level... I don't like doing the same work for a long time. But [my] manager is not able to understand what I'm trying to say. If I have to write and ask about the promotion, they [the management team] will be asking questions... I am afraid that they will give me lecture on it, so its better to not talk about it." – *P6*

Apart from written messages, two participants stated that they took the help of their hearing co-workers often to communicate with their manager. Our participants were more comfortable in having a written exchange with their co-workers, because the co-workers were not as busy as their managers and hence our participants could have a relaxed informal conversation with them. P14 (who is the sole Deaf software engineer in his organization) said the following when asked about whether he and his manager have ever discussed a salary hike:

> "When I need to have some conversation with my manager, I will first talk to a co-worker. Then he/she will come with me to explain to the manager. I used to write [on paper] and explain to co-worker for long conversations. Now I use WhatsApp to communicate [with the co-worker]... Conversations are usually about work-related issues. I never spoke about my salary with my manager." – *P14*

Other participants like P6 reported having limited conversations with his manager:

> "Before I joined [the team], there were Deaf people working already. They taught the manager sign language, but not much. I cannot talk about my job role to my manager. I can only say hi/bye to him. He [my manager] only understands conversations like 'did you have your tea?'." – *P6*

P6 works at an e-commerce warehouse that does not employ an interpreter. His job is to pack and seal products into boxes before they get shipped to customers. Due to his blue-collar work profile, he finds it difficult to communicate over WhatsApp/email.

4.4.5 Conversations with co-workers. Except P8 (who works in an NGO with only Deaf employees) and P13 (who works in a college for the Deaf), all our interview participants work in workplaces where a majority of co-workers are hearing. Nine participants reported working in teams where they were the sole Deaf employee. This results in a need for frequent interactions between the Deaf and hearing co-workers, from casual conversations to formal discussions. However, all participants reported that the frequency and quality of interactions with their hearing colleagues were poor. E.g.,

> "I have only hearing colleagues at work. Conversations with my hearing friends won't go beyond simple greetings like 'Hi', 'Bye', etc. Sometimes the CEO meets me, and when he does he will ask 'How is your work?', 'How are you doing?' etc. During these times I would have communication problems and so I taught a few signs to the CEO." – *P14*

Such interactions happen using simple, intuitive signs that hearing co-workers can understand without any sign language training/knowledge.

In offices with interpreters, the interpreters make meetings accessible. However, they are usually not present for informal, unscheduled water cooler conversations. This results in minimal social interaction for the Deaf employees.

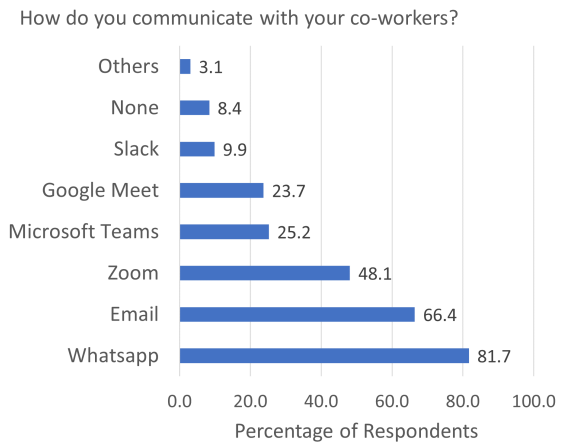

Fig. 3. Responses to the question "How do you communicate with your co-workers at work?". The responses have been normalised by the total number of participants taking part in the survey

Our survey shows that WhatsApp and email are the most common ways to communicate with co-workers. 62.6% of our participants also used video conferencing tools, as they were working from home due to COVID-19. We found that Zoom was most commonly used (48.1% of our participants used it), followed by Microsoft Teams (25.2%) and Google Meet (23.7%). This may reflect their organization's adoption of specific tools, but not necessarily the accessibility of these tools. Coincidentally however, our participants preferred Zoom (even in offices using Teams/Meet), as Zoom allows users to quickly shift between participant video tiles on a call (even on a mobile device), thus enabling Deaf users to focus on the person speaking or signing. Google Meet on the other hand, automatically identifies the speaker and prominently displays the speaker's video on every user's screen by default, which our participants found "*unusable as an interpreter was also on the call*".

4.5 Other Everyday Scenarios

Our survey results showed that the Deaf community face communication challenges in a variety of everyday scenarios. Here, we discuss the scenarios mentioned in Figure 1, using excerpts from our interviews.

4.5.1 Phone Calls. Our participants never make audio calls. However, they occasionally receive audio calls. These calls are generally from food delivery personnel (e.g., Swiggy, Zomato), e-commerce delivery personnel (e.g., Amazon,

Flipkart) or ride-hailing drivers (e.g., Uber, Ola). The calls are of 2 types—confirmatory calls informing the service requester that they have arrived at their doorstep and query calls asking for guidance to reach the exact address of the service requester. Addresses in India are not well-defined, hence such query calls are not uncommon [19].

> "When I book a cab, it is difficult to guide the person to my place. If I go to the doctor for fever or any problem, I can at least write and show them. But with drivers, there is no way to make them understand. Sometimes I text, but they don't reply and keep calling." – *P9*

The query calls cause distress to our participants as they can neither understand nor reply to the callers. Five participants reported taking their hearing neighbours' or co-workers' help to answer such calls on their behalf. The COVID-19 pandemic has resulted in a steep rise in the usage of home delivery services, thus exacerbating this accessibility problem.

4.5.2 Banks and Post Offices. Participants complained that bank and post office officials did not understand sign language, but hiring an interpreter to accompany them while visiting these places is prohibitively expensive. Their only option was to take help from a hearing person. However participants like P8 were hesitant to "*disturb*" their family members and friends for every task. The combination of lack of interpreters, crowded bank, and complex procedures makes it a challenging scenario:

> "I went to the bank to change my ATM card. There was already a queue at the counter, so I did not have the time to write and show to them and explain... It was difficult... I went to the bank in Delhi. They say you have to go to Kerala [a state in South India] and do this work. There was a lot of communication problem with hearing people. They asked me to get someone from my family to assist me. It was my work, but I was not able to communicate with him [the bank official]. Also many people were there, so he can't spend a lot of time with me. I was trying to explain to them that I had filled a form for a new ATM card because I wanted to change it. They were not able to understand me and asked me to come back with my mother." – *P13*

Even commonly used workarounds like writing on a piece of paper do not work, due to the time needed for a two-way written conversation. Participant P3 reported bank managers "*getting angry*" at him for the delays caused in writing-based communication, as he was "*holding up the queue*". Despite these difficulties, P10 reported that her Deaf primary school taught her the kind of questions to ask at banks and similar institutions. With this training, she is able to manage independently.

4.5.3 Home. We found that our interview participants faced challenges in communicating with their own family members. In particular, 7 participants stated that their parents do not know sign language, and that conversations with their parents are limited to simple questions and gestures, like "*Did you eat your food?*". A few participants (like P1) are able to lipread their parents. Parents often discourage the use of sign language due to the stigma associated with it and encourage their children to speak. Such social stigma against people with disabilities has been known to result in significant challenges [54].

> "I was in a boarding school from 1st to 10th [grade], where we had a sign language interpreter... My parents talk a lot. I taught them few signs, but they don't like me signing, so they always speak. If I don't understand then they use sign language. I don't have much conversation with them. My parents tell me to practice speaking even if I don't understand it... I don't like to interact with them much." – *P12*

Three participants were able to communicate with their parents in sign language—two of them have Deaf parents and the third taught sign language to his hearing parents.

Our participants reported that their siblings were more receptive to learning sign language. Four participants have hearing siblings who know "*ISL really well*" and with whom they can have complete conversations. For conversing with their parents, most participants took the help of their siblings (or a friend) who can both sign and speak.

4.5.4 Hospitals. Our participants found visiting a doctor to be accessible, as they are generally "*patient and educated*", and agree to have a written conversation. However, miscommunication with a doctor can be potentially risky as it can lead to misdiagnosis and/or incorrect medication. Hence, the Deaf community prefers to take the help of a hearing family member, friend, or an interpreter for doctor visits, particularly for serious illnesses.

> "Hospitals are good, no problem. The nurse is also there, they help us, they can read. So if it is a small thing like fever then I can write and show. But if I have a major thing, then I have to bring an interpreter for better understanding. I don't know how to simplify the language and explain. It's better to take someone who knows sign language, as it's risky and a big issue if we don't understand what they [the doctor] are saying. For example, what if we go and try to explain to doctor but we still have the doubt of whether we should take the medicine or not. So it is better to take an interpreter or your family member." – *P13*

Moreover, as per P10, the sentence structure and English grammar used by the Deaf community increases the chances of miscommunication in a written conversation.

4.5.5 Transportation. We found that participants use 3 modes of transportation for their daily commute—cabs, auto-rickshaws, and public buses. As discussed in Section 4.5.1, ride-hailing cab drivers call their passengers for confirmation and address-related queries. All our participants have experienced their rides being cancelled by the driver when they did not pick up the driver's phone calls. Attempts by our participants to interact with them over text messages were mostly unsuccessful, as drivers were usually driving when our participants made the phone call and hence were unable to check and reply to messages. This inability to communicate with cab drivers emerged as a strong pain-point with 9 participants complaining about it. The most common solution was to take the help of a nearby hearing person to interact with the driver.

P4 stated a workaround, wherein his organization provides a free pick-up and drop service at a fixed time for its Deaf employees. If the employee does not show up on time for his/her pickup, the driver calls the office interpreter and the interpreter checks with the Deaf employee.

Despite the phone call related challenges, our participants preferred app-based ride-hailing services over hailing auto-rickshaws on the road: "*it is still easier to book a cab or an auto on the app than to catch one on the road as it is difficult to explain the destination to auto drivers. In ride-hailing apps, it is easy to put the [destination] address in text.*" – P1. Moreover, a large fraction of auto-rickshaw drivers do not understand English, which makes written message-based communication infeasible.

Lastly, our participants face unique challenges while using public buses. Public buses are the most affordable means of transportation in India and hence tend to be overcrowded. Moreover, the rider needs to know the correct bus stop to get down at, and the name of the next stop is usually announced by the bus conductor. Thus, a bus ride requires communication with the bus conductor or co-passengers, which is difficult using written messages, both due to the crowd and co-passengers often being unable to read English. For instance:

> "While travelling in the bus when I ask 'Which place is this?', they cannot understand and sometimes ignore me. When I write and ask them, people don't understand English because they are from village. That's why I have to write in Tamil and show them the paper." – *P7*

4.5.6 Restaurants. Restaurants do not pose a challenge for our participants, as conversations are usually short and happen on print (through the menu card and bill). Written message exchange is also feasible due to the asynchronous nature of interaction, i.e., the Deaf can write before the waiter arrives for the next task.

> "We will have the menu card. I'll show them like this *[points at menu card]*. And whatever is the money, I pay them. I've never faced problem with communicating. I go alone and also along with my friends." – *P15*

4.5.7 Shops. We inquired about 2 types of in-person shopping—frequent purchases such as buying groceries and occasional purchases such as shopping for clothes. Our participants preferred visiting large supermarkets over small grocery stores, as these supermarkets permit self-service, with prices listed on all products and aisles clearly labelled with their product categories.

In contrast, for clothes shopping, our participants preferred local smaller cloth markets due to the cheaper prices. Cloth markets have multiple shops selling nearly identical items. Participants were comfortable shopping there as their interactions were limited to asking for the price. Moreover, the presence of multiple shops meant that if a storekeeper was unable to communicate well with them, they could move to the next shop. For instance:

> "I never used to go shopping as my mother always bought clothes for me... After coming to Hyderabad [where I live alone], I started going shopping with [Deaf] friends. Now, I buy my own dresses. If I do this sign *[does the sign for money]*, the shopkeeper will show their fingers *[holds up fingers to indicate the price of the product]*. If the price is high, I bargain. I will try to write down my price and negotiate. If they don't understand, I will just leave and go the next shop. I don't face any problems." – *P15*

5 DISCUSSION

Our survey and interviews highlight key challenges faced by the Indian DHH community on a day-to-day basis. The community uses a variety of workarounds to tackle these challenges. In this section, we examine key findings in the broader context of existing work and provide design recommendations.

5.1 Agency

In an effort to gain agency and avoid stigma, our Deaf participants were encouraged by their teachers and parents from early childhood to speak, lip-read, use hearing aids, take speech therapy, and avoid signing in public. In fact, these methods were prevalent in schools for the Deaf, with 49.6% of our survey respondents stating that they had undergone speech therapy. While such approaches worked with moderate success for two hard of hearing participants, all profoundly deaf participants were unable to and/or uncomfortable with using these methods. Thus, well-meaning efforts by members of the hearing community to increase the agency of Deaf individuals by reducing stigma actually have the opposite effect. The role of stigma in influencing decision-making amongst the guardians of children with disabilities has been previously reported, particularly in the Indian Blind population [45]. They found that parents did not provide white canes to their Blind children to avoid stigma and make their children '*look less blind*' [45], which curtailed their children's physical and mental growth. This suggests that social stigma supersedes agency and independence in India, and serves as an additional hurdle for people with disabilities.

We found that the agency of a Deaf individual varies across scenarios. A few factors influencing agency are the nature of interaction, the time available for communication, and the English comprehension ability of the hearing person. For activities like shopping and ordering food at restaurants, we found our participants independently interacting with the hearing person using a combination of basic signs and written messages as the required interaction is short and there is time available. These interactions may fail if the hearing individual lacks the ability to comprehend English. Our participants face difficulties in independently communicating with a driver or a doctor, as drivers usually have no time to have a written conversation, and conversations with a doctor have a higher risk of misinterpretation. In such scenarios, the Deaf took the help of an interpreter or another hearing person (friend, family or co-worker). However, this reduces their agency and raises privacy concerns. Privacy may not be a major concern for short conversations with strangers (such as with a driver or a shopkeeper), but can be more relevant when conversing with a manager at work or while interacting with a doctor about a stigmatized health condition.

5.2 Workplace Accessibility for DHH employees: India versus the US

Foster et al. [36] detail challenges faced by the American Deaf population at the workplace, focusing on Deaf supervisors. In contrast, none of our 15 interview participants and only 1 (out of 131) survey respondents were in a supervisory/managerial role. This suggests that roles requiring stronger communication skills are not common for the DHH community in India. This may be be attributed to the lack of resources in the Global South, for instance in providing access to full-time interpreters. The role of the interpreter is in sharp contrast across the Global North and South—in the US, most interpreters were secretaries of Deaf employees [36], while our participants reported that most interpreters were managers or colleagues. This suggests a power imbalance which may result in limited agency of the Deaf employees in India. With respect to technology, as early as 2003, the deaf employees in the US had access to relay services and TDDs (Telecommunications Device for the Deaf) [36]. TDD is an electronic device for text communication over a telephone line. However, none of our participants reported having access to comparable service/technology even in 2021. In our interviews, we found our participants relying minimally on technology, primarily limited to WhatsApp video calls and messages. When communicating with co-workers, we found similarities between Indian and US Deaf employees with respect to the usage of written notes and emails for crucial messages, and minimal informal conversation or chit-chat. Lip-reading was found to be commonplace in the US, while in India, only one participant knew lip-reading perhaps due to the large number of spoken languages in India. With respect to signing, a few hearing co-workers in the US had even learnt ASL, which is in contrast to India where none of the participants reported colleagues who formally learnt ISL. This may be again due to the difference in job seniority of the Deaf employees in US and India. Please refer to Table 2 for a summary of these similarities and differences.

5.3 Designing Inclusive Technology

Interpreters on Demand: Using an interpreter emerged as the most reliable solution for the Deaf community. All of our participants mentioned the ease of interacting with hearing individuals in the presence of an interpreter, and mentioned a reduction in anxiety during such conversations. However, accessing interpreters at all times is not feasible due to the high cost of hiring an interpreter, the dearth of certified interpreters in India, and the inability of interpreters to be physically present on demand. Technologies like Video Relay Services solve these problems. Using a relay service has multiple advantages–it is cheaper, enables an interpreter to serve more people (as they can switch calls), and allows Deaf users to hire on demand. While this technology is well developed in the Global North, it is not available in India.

Table 2. A comparison of our key findings at the workplace with a similar study conducted in the United States.

Result	Findings from Foster et al. [36]	Our Findings
Job Role	In supervisory roles at work.	In internal contributor roles.
Interpreters	Easy access to interpreters.	Minimal access to interpreters. Limited to meetings.
Interpreter Position	Interpreters are a secretary to the Deaf employee in most cases.	Interpreters are a manager of the Deaf employee in most cases.
Phone Calls	Access to TDD and relay services.	No support to make phone calls accessible.
Lip Reading	Used lip-reading to converse with hearing people.	Only one participant could lip-read.
Writing	Emails and written notes.	WhatsApp, emails and written notes.
Chit-chat	Left out of informal conversations at work.	Left out of informal conversations at work.
Signing	A few co-workers learnt sign language to communicate with their Deaf manager.	Signing was limited to simple signs which did not require formal sign language knowledge.

The closest to a video relay service we found is an app called SignAble[7] which was recently launched in 2020 in India. However, none of our participants were aware of it. We identify a strong requirement for a video relay service in a variety of scenarios, and believe that such an app for the Indian context would be potentially useful if effectively advertised to the Indian DHH community.

Video/Audio Calling Tools: Video calling was used often by our participants in multiple scenarios, with an increase in usage due to COVID-19. The applications used for video calling varied—WhatsApp is used with friends and family, and Zoom, Google Meet, and Teams are used at work. In these applications, speech-to-text (STT) technology is used to auto-generate captions and serves as a partial technological alternative to an interpreter, which our participants found useful for work meetings. However, they experienced two issues while using the captions: (a) inaccuracy in generated captions as the STT tools often were inaccurate in understanding speech with Indian accents, and (b) their inability to read the generated captions at the required pace due to their limited proficiency with English. A simple solution to address this would be to allow a history of the caption that the Deaf people can revisit. A more technically challenging solution would be to integrate automatic tools for text simplification [74] such as Lexi [20]. Another challenge with video conferencing tools is the use of the screen real-estate. Tools like Google Meet default the primary video on each user's screen to the current speaker based on audio signals. This design choice makes the tool inaccessible to the DHH community, who would like to see the interpreter prominently at all times. This was also reported as a challenge when viewing shared content such as slide decks. Finally, this accessibility challenge also manifests in recordings of meetings where the content recorded depends on automatic choices made by the tools which are not necessarily informed by accessibility for the Deaf. To address these challenges, video conferencing apps can provide more agency to Deaf users by allowing manual configurations for what is laid out on the screen, and can let Deaf users decide what fraction of the screen should be used for each content. Apart from the video calling applications, a few apps on the Google Play Store (such as Rogervoice[8]) attempt to make audio phone calls accessible for the DHH community by extensively using STT and TTS (text-to-speech) technologies. Rogervoice auto-generate captions for incoming audio calls, and provides an optional pre-recorded message to inform the caller that they are speaking to a Deaf person. The Deaf person can either talk to the caller or type out a message that is read to the caller using TTS technology. Unfortunately, none of these apps are available in India.

[7]https://play.google.com/store/apps/details?id=org.signable.apprtc
[8]https://play.google.com/store/apps/details?id=com.rogervoice.app

Ambient Conversation: Our participants struggled to take part in informal social conversations with co-workers. Similar challenges were reported among DHH employees in the US [36]. Even listening to ongoing conversations while not contributing can help DHH employees connect with their co-workers. Certain apps exist that enable Deaf users to listen to ambient conversations. For instance, Talk to Deaf[9] uses Google's STT technology to display captions for conversations happening around the user's smartphone. However, we found that the app worked only when a person is speaking directly into the microphone and when there is minimal ambient noise, severely limiting their utility. Improved microphone technology for indoor settings is available on many consumer-grade smart speaker devices which can add support to help the DHH hear ambient conversations.

Accessible Service Apps: While efforts to build applications that improve accessibility continue, several problems faced by the Deaf community can perhaps be solved by designing existing applications more inclusively. In particular, a majority of our participants faced challenges in interacting with cab drivers and e-commerce/food delivery personnel over phone calls, and have to seek help from family, friends, neighbors, co-workers or strangers. These service apps may add a "Do not disturb mode" or "DHH mode" informing the driver/delivery person that the service requester is a Deaf individual. In such cases, communication can be over messages instead of phone calls.

"Broken" English: The underlying difference in sentence structure between English and ISL and the limited English proficiency of the Deaf community results in Indian DHH members writing in a form that may be incomprehensible to hearing people. A tool that inputs text in ISL-like SOV structure and generates text in SVO structure with appropriate grammar checks may address this accessibility gap. Moreover, the output text can be translated to local languages using existing Neural Machine Translators (NMTs), potentially resolving challenges faced by the Deaf as reported while using public transportation.

Conversion between ISL and text: Automatically converting sign language videos, as captured by a smartphone camera, to natural langauge text is a challenging task. However, this is an active area of research in the AI community and several recent papers have made progress [21, 29, 41, 77]. Like other AI tools, training a sign language recognition system requires large amounts of training data, which consists of sign videos and their corresponding text translations. Given its diversity, a large and concerted effort is required to create such datasets for ISL. Further, such data collection effort may initially be focused on the most sensitive contexts, such as communication at the workplace, with service providers, at banks, and while travelling. Finally, such datasets must be released in the open-source to enable AI researchers to build sign language recognition systems. Integrating such a tool would be most useful in communication apps like WhatApp, which is used by 95.7% of survey respondents for personal calls and 81.7% of them for professional calls. This would require that the sign language recognition tool be compatible with low-to-medium end smartphone cameras.

6 CONCLUSION

In this work, we study the challenges faced by the Deaf community through interviews with 15 Deaf participants and 131 Deaf survey respondents. Our study reveals the various challenges faced by the Indian Deaf community on a day-to-day basis. First, we discuss foundational challenges due to the stigma associated with signing in India, the English literacy issues faced by the Deaf, and the problems arising due to the variations in ISL, which is characteristically different from other sign languages like ASL. Next, we cover context-specific challenges such as the lack of interpreters in the workplace, the inability of the Deaf population to converse with managers and hearing co-workers, and accessibility

[9] https://play.google.com/store/apps/details?id=unique2040.com.text2speech

challenges in other common places such as the home, banks and hospitals. Lastly, we cover technology related challenges, where we highlight problems with video calling applications and automated captioning services, and discuss the issues with app-based service delivery. We end by contrasting the Indian workplace challenges with the experiences of Deaf employees in the Global North, and propose technology and design recommendations to tackle the three types of identified challenges.

7 ACKNOWLEDGEMENTS

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